

Lab 1 – Ideal Gas Law

CHE 260 – Thermodynamics and Heat Transfer

Group Members

Chance Hattrick	Student ID
Ammar Hasham	1011941124
Muhammad Hasnain Heryani	1011884695

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1 Introduction

2 Results and Discussion

2.1 Part 1: Path Independence and Tank Volume Ratio

Part 1 of the lab discusses the comparison between the movement of air between two identical volume tanks, one with a gauge pressure of 40 PSI to another with a gauge pressure of 0-PSI. The main point of comparison is the differences in a slow equalization of pressure versus a rapid equalization of pressure.

2.1.1 Processed Results: Rapid and Slow Expansion

Figures 1–2 show the measured pressure and temperature histories for the rapid and slow expansions.

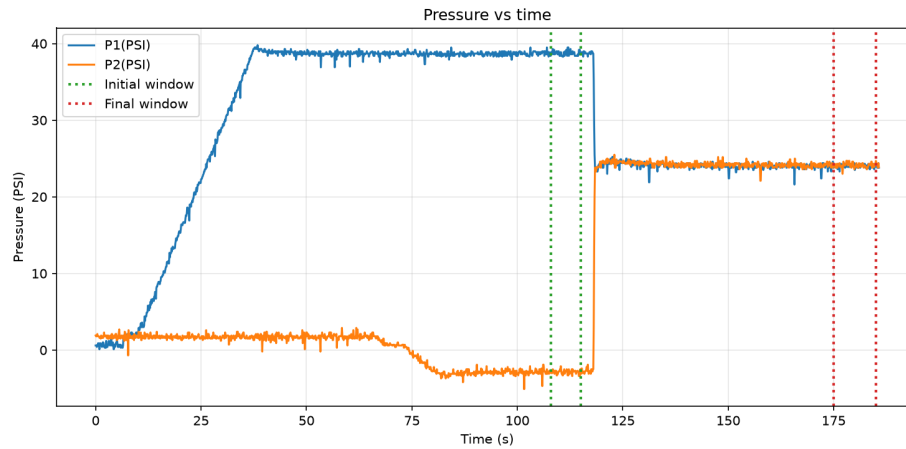


Figure 1: Tank pressures during the Part 1a rapid expansion.

Taking averages across the marked time ranges on Figures 1–2 for final pressure and temperature values, we obtain the following values:

$$P_{2R} = 24.13 \text{ PSI}, P_{2L} = 24.03 \text{ PSI}$$

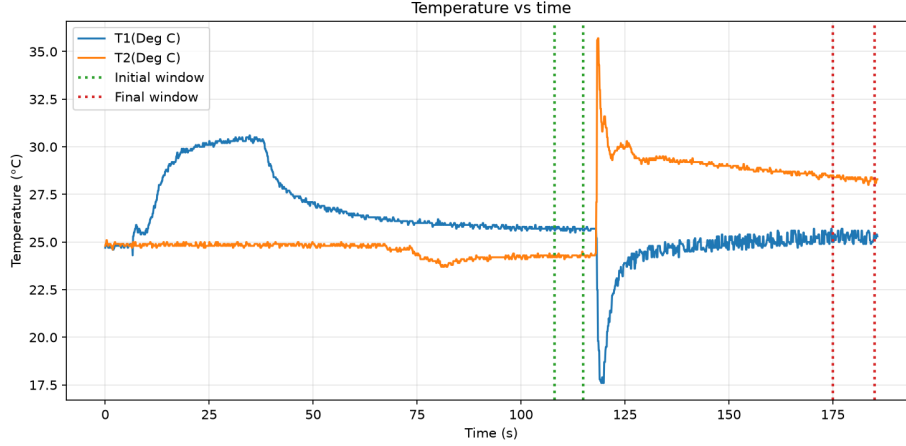


Figure 2: Tank temperatures during the Part 1a rapid expansion.

$$T_{2R} = 28.33^{\circ}\text{C}, T_{2L} = 25.25^{\circ}\text{C}$$

Using these values with Ideal Gas Law, we can determine the proportionality of pressure, volume and temperature between the two tanks at the final state. It can be assumed that the ratio of the final pressure, volume and temperature of the left tank is equal to the final pressure, volume and temperature of the right tank since both tanks are in equilibrium with each other at the final values. The following calculations show the resultant volume ratio of the rapid equalization of pressure

$$\begin{aligned}\frac{P_{2L}V_L}{T_{2L}} &= \frac{P_{2R}V_R}{T_{2R}} \\ \frac{V_L}{V_R} &= \frac{P_{2R}}{P_{2L}} \cdot \frac{T_{2L}}{T_{2R}} \\ \frac{V_L}{V_R} &= \frac{24.13}{24.03} \cdot \frac{25.25}{28.33} \\ \frac{V_L}{V_R} &= 0.895\end{aligned}$$

Similar to Figures 1–2, the following Figures 3–4 show the pressures and temperatures of both the left and right tank during the slow equalization of pressure part of the lab.

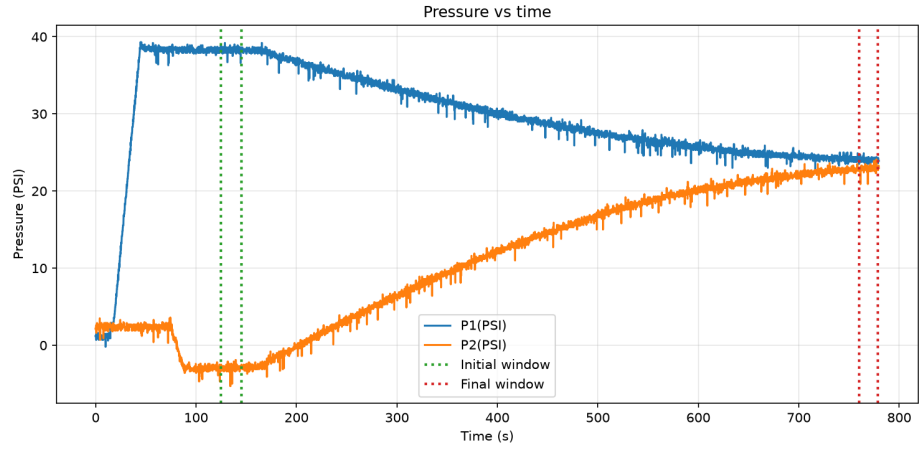


Figure 3: Tank pressures during the Part 1b slow expansion.

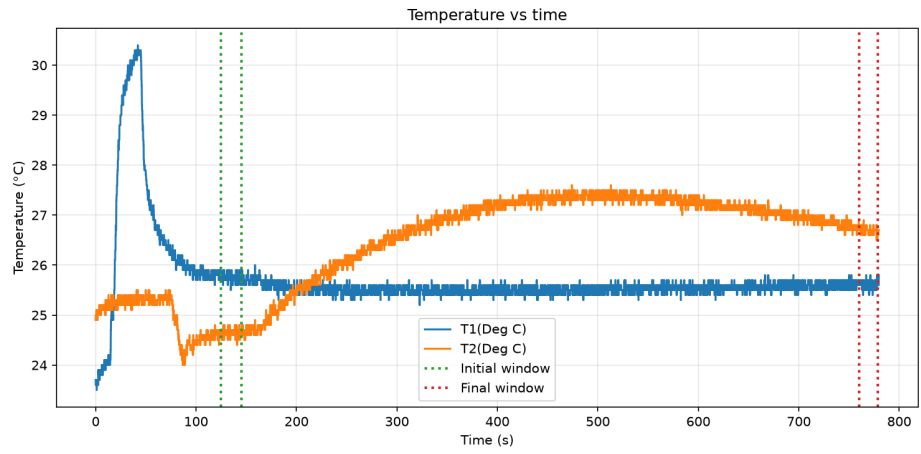


Figure 4: Tank temperatures during the Part 1b slow expansion.

Taking averages of the final pressures and temperatures at the marked time ranges on the graphs, similar to Figures 1 and 2, we obtain the following values:

$$P_{2R} = 22.94 \text{ PSI}, P_{2L} = 24.04 \text{ PSI}$$

$$T_{2R} = 26.70^\circ\text{C}, T_{2L} = 25.60^\circ\text{C}$$

Then applying the Ideal Gas Law in the same way, we obtain the following:

$$\begin{aligned}\frac{V_L}{V_R} &= \frac{P_{2R}}{P_{2L}} \cdot \frac{T_{2L}}{T_{2R}} \\ \frac{V_L}{V_R} &= \frac{22.94}{24.04} \cdot \frac{25.60}{26.70} \\ \frac{V_L}{V_R} &= 0.915\end{aligned}$$

2.1.2 Tank Volume Ratio

2.1.3 Heat Transfer and Net Heat Transfer

2.1.4 Path Independence of State Properties

2.2 Part 2: Initial Mass and Volume of the Left Tank

2.2.1 Processed Pressure, Temperature, and Mass Flow Results

2.2.2 Initial Mass and Tank Volume

Knowing that all added mass is measured by the sensor, we can calculate the added mass in the left tank as the integral of the mass flow rate - time graph. Using numerical methods, we can find that the total mass inside the tank is 23.259g.

Since the mass inside the system after any time is known, we can create a system of

equations to calculate V and initial mass.

$$PV = (m_i + \delta m)RT$$

$$P_1 =$$

2.2.3 Sources of Error and Limitations

There are several key sources of errors and limitations on this section of the lab, either relating to random human error or systematic error with the apparatus.

When setting the pressure of the system to 40 PSI, we were reliant on a teammate manually checking the gauge pressure. This is limited by human reaction time and uncertainty in the gauge pressure reading, both due to parallax error and uncertainty in measurements. Thus, there was a lot of error with setting the initial pressure to exactly 40 PSI. This is seen through the sensor data which showed that the pressure was slightly below our target of 40 PSI.

Additionally, sensor data displayed that the mass flow rate was not completely constant at 50 g/s, even when temperature and pressure have stabilized for the air in the tank.

2.2.4 Effect of Air Compressibility

The compressibility of air can greatly affect experimental results of pressure and volume, especially as experimental conditions diverge from assumptions made by the ideal gas equation. The ideal gas equation assumes that temperatures are kept high and pressures are kept low. When pressures are kept low, intermolecular forces can be assumed to be negligible. With this model, the compressibility of air can be modelled to be constant. As the pressure increases, electromagnetic repulsions and attractions between particles greatly contribute to the behavior of the gas. This causes the linear model of air compressibility to break down, displaying the inaccuracies in the ideal gas model. Since the intermolecular forces

increase as pressure increases, our calculated compressibility is much higher than the actual compressibility.

One can model this using the compressibility factor, where

$$Z = \frac{Pv}{RT}$$

or

$$PV = ZRT.$$

Under constant volume, this measures the divergence of a physical model from Gay-Lussac's Law.

During our experimentation, the maximum gauge pressure found was around 40 psi. Given that this is a similar gauge pressure to that seen in pressurized car tire, this is high enough where our apparatus diverges from assumptions made by the ideal gas model and we should check the compressibility factor during our process.

If we hold the volume V constant, we can calculate the compressibility factors at low and high pressures.

At $t = 10$:

$$1.3 \cdot v = Z_1 \cdot R \cdot 317.85,$$

$$Z_1 = \frac{1.3 \cdot v}{317.85 \cdot R} = \frac{2}{489} \cdot \frac{v}{R}.$$

At $t = 50$:

$$52.2 \cdot v = Z_2 \cdot R \cdot 301.45,$$

$$Z_1 = \frac{52.2 \cdot v}{301.45 \cdot R} = \frac{1050}{6029} \cdot \frac{v}{R}.$$

Thus, through these calculations, we can see that the compressibility of air signifi-

cantly increases at higher pressures.

3 Conclusion

References